GENERATIVE MUSEOLOGY AND KINETIC SYSTEMS

Thinking Cells Frozen and in Motion

Botanical diffusion and kinetic choreography for museum collections.

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ABSTRACT

Digital museum catalogs often present holdings as static thumbnail tables or isolated three-dimensional meshes that strip objects of relational context. Instead, through *Thinking Cells Frozen and in Motion*, a client-side web installation built with twenty-five photographs from the MIT Museum collection, we aim to recreate and reanimate the cognitive and mechanical effort of thinking machines. In view, each specimen (cell) remains *frozen* as an archival photographic record while simultaneously entering *motion* through morphological shaders, an edge-conditioned ONNX diffusion network, and planar orbital choreography that stage a visual dialogue between biological growth and engineered instruments.

Museums of science and technology steward artifacts that capture physical processes frozen at a decisive instant, whether a microsecond strobe flash by Harold Edgerton, annual timber rings in a cypress core, or cursive notes in a laboratory log. When published online, standard search grids isolate each record in a static cell, whereas three-dimensional scanners favor metallic hardware over textiles, paper, and biological media.

Building on Whitelaw's concept of generous interfaces [1], we ask how a flat collection grid behaves when every tile operates as a coupled visual cell. Rather than synthesizing ungrounded imagery from text prompts, our system anchors every operation in verified collection photographs that define spatial boundary conditions for reaction-diffusion waves [2, 3], curl-noise advection [4], and client-side denoising diffusion [5, 6, 7].

Crucially, the interactive workbench pairs MIT Museum artifacts with an eight megabyte U-Net trained on historic botanical drawings [6]. Conditioned on the binary Sobel silhouette of a carved pine hull or an interferometer image, the network projects organic stippling and foliate veins across engineered forms. This cross-domain mismatch surfaces structural homologies between natural morphogenesis and scientific instruments while periodically dissolving back to the unretouched photograph.

01 DESIGN PRINCIPLES

- **Archival Grounding:** Twenty-five MIT Museum photographs spanning 1888 to 2025 represent fiber, wood, pulp paper, glass emulsion, agar cultures, and brass instruments.
- **Frozen and Kinetic Duality:** Cells alternate between unretouched photographic clarity and active morphological synthesis so filtering never obscures the primary source.
- **Botanical Cross-Reading:** Edge-conditioned inference projects a botanical drawing prior onto technical artifacts, inviting visitors to compare organic and mechanical line structures.
- **Tactile Hardware Hierarchy:** Header controls consolidate into cellular modes (`etch`, `reaction`, `flow`, `halftone`), a collapsible `controls` drawer cued by a softly glowing green hand indicator, and a `light` / `dark` switch.

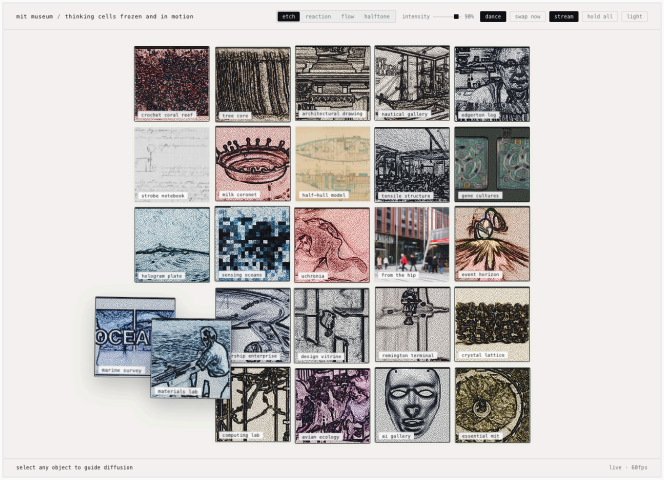


Figure 1. Light mode gallery view (`#f5f4f0` paper ground). In the lower left, the *marine survey* and *materials lab* cells elevate and curve around each other along opposite normal vectors while adjacent cells render Atkinson stipple plates.

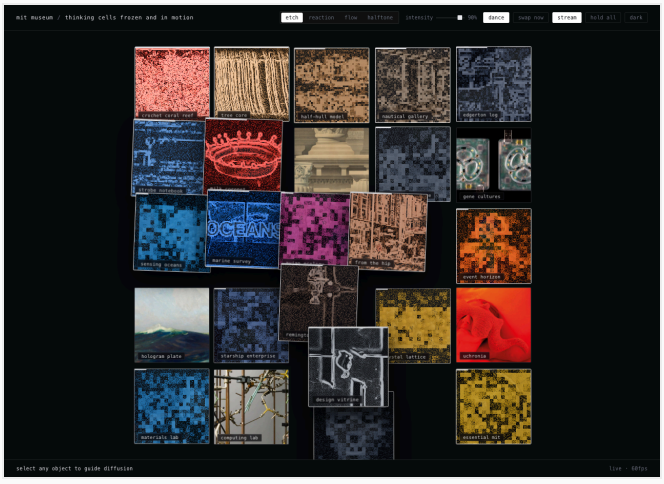


Figure 2. Dark mode gallery state during a multi-cell ensemble swap, showing coarse-to-fine block denoising converging onto Sobel contours across twenty-five collection specimens.

EDGE-CONDITIONED MULTI-SCALE U-NET OPERATOR ($F_\Theta : \mathbb{R}^{13\times64\times64} \to \mathbb{R}^{3\times64\times64}$)

$$\mathbf{z}_{\text{in}} = x_t \oplus x_{\text{low}} \oplus \mathbf{u} \oplus E_{\text{sil}} \oplus \sigma_t \mathbf{1}_{64\times64} \in \mathbb{R}^{(3+3+3+3+1)\times64\times64}$$

$$h_0 = \Phi_{\text{stem}}(\mathbf{z}_{\text{in}}), \quad h_\ell = \mathcal{E}_\ell(h_{\ell-1}\downarrow_2), \quad \tilde{h}_{\ell-1} = \mathcal{S}_{2\times2}\Big(\mathcal{D}_\ell(\tilde{h}_\ell \oplus h_\ell)\Big), \quad \hat{x}_0 = \Psi_{\text{out}}\Big(\tilde{h}_0 \oplus h_0\Big)$$

Mathematical Model. At each denoising step, the five conditioning fields—noisy patch x_t, bilinearly upsampled parent patch x_{low}, normalized coordinates $\mathbf{u} = (u, v, 0.5)$, binary Sobel silhouette E_{sil}, and noise level σ_t—concatenate along channels ($\oplus$) into $\mathbf{z}_{\text{in}} \in \mathbb{R}^{13\times64\times64}$. The stem Φ_{stem} lifts $\mathbf{z}_{\text{in}}$ to 64 channels, four strided encoder stages $\mathcal{E}_\ell$ ($\ell \in \{1,2,3,4\}$) contract spatial resolution across $64^2 \to 32^2 \to 16^2 \to 8^2 \to 4^2$ with dense-concatenated residual projections ($\mathbb{R}^{192} \to \mathbb{R}^{64}$), and four decoder stages $\mathcal{D}_\ell$ fuse skip features ($\oplus$) before applying the 2×2 depth-to-space PixelShuffle operator $\mathcal{S}_{2\times2} : \mathbb{R}^{256\times H\times W} \to \mathbb{R}^{64\times2H\times2W}$ and output projection Ψ_{out} to recover $\hat{x}_0 \in \mathbb{R}^{3\times64\times64}$.

02 SYSTEM ARCHITECTURE

2.1 Offline Corpus Packaging

When a web application is served over HTTP, standard image tags load cleanly into HTML5 canvases. However, museum gallery kiosks and archival review bundles are frequently launched directly from local storage via the `file://` protocol, where browsers assign each relative image file an opaque origin and block `getImageData()` reads.

To support zero-configuration offline execution alongside standard web hosting, a Python preprocessing script center-crops each photograph to 384×384 pixels with a Lanczos filter and serializes the corpus into `museum_corpus_data.js`. While base64 encoding increases the bundle size by roughly thirty-three percent to 1.09 MB, data URIs inherit the document security context and allow synchronous pixel ingestion in 45.10 ms across all twenty-five cells.

2.2 Structural Feature Tensors

At startup, each cell downsamples its image into a 160×160 working buffer ($N = 25,600$ pixels) and computes four structural fields:

1. Affine-Normalized Luminance: Perceptual luminance $L_0 = (0.299R + 0.587G + 0.114B)/255$ is normalized across each specimen's dynamic range $[L_{\min}, L_{\max}]$:

$$L(x,y) = \text{clamp}\left(\frac{L_0(x,y) - L_{\min}}{\max(0.15,\, L_{\max} - L_{\min})},\, 0,\, 1\right)$$

2. Sobel Gradient Field: Convolving $L(x,y)$ with 3×3 Sobel kernels yields horizontal and vertical derivatives (G_x, G_y) and clamped edge magnitude:

$$E(x,y) = \min\left(1.0,\, 1.45\sqrt{G_x(x,y)^2 + G_y(x,y)^2}\right)$$

3. High-Pass Engraving Plate: Subtracting the 5×5 strided local mean luminance $\bar{L}_5(x,y)$ isolates surface relief regardless of uneven vitrine lighting:

$$H(x,y) = \text{clamp}(0.28 + 2.4(L - \bar{L}_5) + 0.85\,E,\, 0,\, 1)$$

Quantizing $H(x,y)$ at threshold 0.48 via Atkinson error diffusion propagates one-eighth of the residual error to six neighboring pixels, preserving crisp midtone stippling across dark backgrounds.

2.3 Diffusion Workbench and Cellular Modes

Selecting any cell opens the interactive workbench, which runs Ollin Boer Bohan's *Dino Diffusion* U-Net [6] adapted from Golan Levin's p5.js implementation [7]. Conditioned on noisy latents x_t, scalar noise level σ_t, low-resolution parent pixels, normalized spatial coordinates, and a 64×64 guidance map, the network predicts a denoised patch $\hat{x}_0 = f_\theta(\mathbf{z}_{\text{in}})$ following a rational schedule with shape constant $\kappa = 0.2$ over normalized step progress $u \in [0,1]$:

$$\sigma(u) = \frac{(1-u)(1+\kappa)}{(1-u)+\kappa}$$

$$x_{t+1} = \frac{\sigma_{\text{out}}}{\sigma_{\text{in}}} x_t + \left(1 - \frac{\sigma_{\text{out}}}{\sigma_{\text{in}}}\right)\hat{x}_0$$

Three modifications adapt this pipeline for responsive curatorial exploration:

1. **Silhouette Conditioning:** Rather than passing raw grayscale photographs that yield muddy latents, the workbench extracts a binary 64×64 contour map combining Sobel edges and deep shadows.
2. **Frame-Batched Inference:** Replacing timer-throttled callbacks (`setTimeout` at 8 ms) with three WebGL inference steps per animation frame reduces base 64×64 synthesis latency from 380 ms to 118 ms while streaming intermediate latents live.
3. **Coupled Cellular Shaders:** In the main grid, a 60 Hz loop drives four selectable modes blended with the original photograph by intensity $\alpha \in [0.15, 1.0]$:
   - *Etch:* Coarse-to-fine block sampling ($b \in \{8,4,2,1\}$) converges onto the Atkinson plate and Sobel contours.
   - *Reaction:* An anisotropic Gray-Scott solver [3] with diffusion rates $(D_u, D_v) = (0.18, 0.09)$, decay rate $k = 0.061$, and edge-modulated feed rate $f(x,y) = 0.038 + 0.018\,E(x,y)$:

     $$\partial_t U = D_u \nabla^2 U - UV^2 + f(x,y)(1-U)$$
     $$\partial_t V = D_v \nabla^2 V + UV^2 - (f(x,y)+k)V$$

   - *Flow:* Advection along incompressible isophote velocity vectors $\vec{v} = (-G_y, G_x)$ [4].
   - *Halftone:* Bi-directional cross-hatching modulated by $L(x,y)$.

2.4 Planar Orbital Choreography

To place specimens in shifting visual dialogue without triggering browser layout reflows, all twenty-five cards use absolute coordinates and GPU-composited `translate3d` transforms. A permutation array maps grid slots to cell indices and schedules non-overlapping cycles of length $m \in \{2, 3, 4\}$.

During a swap from $\vec{p}_{\text{from}}$ to $\vec{p}_{\text{to}}$ with displacement $\Delta\vec{p} = (\Delta x, \Delta y)$ and cubic easing $s(u) \in [0, 1]$, each card deflects along the unnormalized perpendicular normal $\vec{n}_\perp = (-\Delta y, \Delta x)$ scaled by lateral factor $\gamma \in [0.28, 0.42]$:

$$\vec{p}(u) = \vec{p}_{\text{from}} + s(u)\,\Delta\vec{p} + \gamma \sin(\pi s(u))\,\vec{n}_\perp$$

Because $\vec{n}_\perp$ scales with $\|\Delta\vec{p}\|$, partner tiles automatically bow outward by $2\gamma\|\Delta\vec{p}\|$ at their midpoint ($s = 0.5$), preventing overlap within an active pair while elevating via scale factor $1 + 0.065 \sin(\pi u)$.

03 IMPLEMENTATION BLUEPRINT

Listings 1 and 2 provide the standalone corpus encoder and the per-frame orbital kinematics update routine.

LISTING 1. PYTHON CORPUS ENCODER — BUILD_CORPUS.PY

```
import io, json, base64
from PIL import Image, ImageOps

corpus = {}
for name in specimen_files:
    with Image.open(f"assets/{name}") as im:
        rgb = im.convert("RGB")
        sq = ImageOps.fit(
            rgb, (384, 384), Image.Resampling.LANCZOS
        )
        buf = io.BytesIO()
        sq.save(buf, format="JPEG", quality=86)
        b64 = base64.b64encode(buf.getvalue())
        uri = "" + b64.decode()
        corpus[f"assets/{name}"] = uri

with open("assets/museum_corpus_data.js", "w") as f:
    body = json.dumps(corpus)
    f.write(f"window.MUSEUM_IMAGE_DATA = {body};\n")
```

LISTING 2. PLANAR ORBITAL KINEMATICS — CHOREOGRAPHY.JS

```
function updateCellKinematics(cell) {
  if (cell.isSwapping) {
    const nextP = cell.swapProgress + cell.swapSpeed;
    const p = Math.min(1, nextP);
    cell.swapProgress = p;
    const s = p < 0.5
      ? 4 * p * p * p
      : 1 - Math.pow(-2 * p + 2, 3) / 2;
    const dx = cell.toX - cell.fromX;
    const dy = cell.toY - cell.fromY;
    const arc = Math.sin(Math.PI * s) * cell.arcSign;
    cell.x = cell.fromX + dx * s - dy * arc;
    cell.y = cell.fromY + dy * s + dx * arc;
    if (p === 1) cell.isSwapping = false;
  }
  const w = Math.sin(Math.PI * cell.swapProgress);
  const lift = cell.isSwapping ? 1 + 0.065 * w : 1;
  const xPx = cell.x.toFixed(1);
  const yPx = cell.y.toFixed(1);
  cell.card.style.transform =
    `translate3d(${xPx}px, ${yPx}px, 0) scale(${lift})`;
}
```

SUBSYSTEM	RESOLUTION	MEDIAN TIME
Feature Ingestion	160×160 × 25	45.10 ms (once)
Multi-Scale Etch	160×160 / cell	0.60 ms / cell
Gray-Scott Solver	160×160 / cell	0.70 ms / cell
Curl-Noise Flow	160×160 / cell	1.40 ms / cell
ONNX WASM Step	64×64 patch	58.90 ms / step
Orbital Kinematics	25 DOM Nodes	< 0.01 ms / frame

Table 1. Measured execution medians in Google Chrome 156 on Apple Silicon (M1 Pro) using `performance.now()` across 25 authentic museum specimens.

04 DISCUSSION AND EXTENSIONS

Curatorial Legibility versus Visual Intensity: High filter mix values ($\alpha > 0.85$) foreground morphological texture at the expense of fine archival inscriptions. Our staged lifecycle addresses this tension by holding the diffused state for three seconds before smoothly interpolating $\alpha \to 0$, returning each cell to its unretouched photograph. Future deployments should pair this system with formal visitor studies and curator interviews to evaluate dwell time and object discovery against static catalog baselines.

WebGPU Compute and Institutional Distillation: Porting the Gray-Scott and curl-noise kernels to WebGPU compute shaders will remove CPU-to-canvas upload overhead across larger collection matrices. In parallel, distilling compact eight megabyte ONNX U-Net weights directly on MIT Museum subsets, such as Hart Nautical ship plans or Edgerton strobe sequences, will allow visitors to switch between botanical, naval, and oscillographic priors.

IMAGE CREDITS AND ARTIFACT AVAILABILITY

Archival and exhibition photographs courtesy of the MIT Museum (Harold E. Edgerton Collection, Hart Nautical Collection, and MIT Museum Galleries, `mitmuseum.mit.edu`), used for non-commercial academic research.

- Live web installation: `https://letters-mitmuseum-brain.web.app`
- Open-source repository: `https://github.com/eoduniyi/mit-museum-brain`
- Corpus & U-Net: `assets/museum_corpus_data.js`, `assets/network.onnx`

05 REFERENCES

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